CONSISTENCY-REGULARIZED MULTI-MODAL 3D OBJECT DETECTION FOR AUTONOMOUS DRIVING VIA BEAM-WISE MIXING AND CROSS-SENSOR KNOWLEDGE DISTILLATION

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FINAL YEAR PROJECT REPORT

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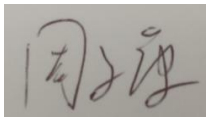
DECLARATION

I hereby declare that this project report is based on my original work except for citations and quotations which have been duly acknowledged. I also declare that it has not been previously and concurrently submitted for any other degree or award at Xiamen University Malaysia or other institutions.

Signature : 

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ABSTRACT

In the field of 3D object detection, model complexity and detection accuracy have always been mutually exclusive. To address this dilemma, This study proposed a Teacher-Student architecture distillation training framework. This framework consists of three main parts: 1. Cross-Architecture Knowledge Distillation; 2. Feature-Level Beam-Wise Mixing; and 3. Consistency Regularization. Experimental results on the nuScenes dataset show that the training strategy of this architecture improves the NDS score of PointPillar from 0.4951 to 0.5468 and the mAP score from 0.3461 to 0.4070. This demonstrates the effectiveness of distillation and BWM perturbation in improving the performance of simple architecture models.

KEYWORDS: Knowledge Distillation, Autonomous Driving, 3D Object Detection, LiDAR, PointPillars.

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CHAPTER 1

INTRODUCTION

1.1 Research Background

What is the core task of autonomous driving perception? Humans have two eyes. They identify objects by the images they see and the prior knowledge in their brains. They then estimate depth by the focusing of their eyes and, similarly, the prior knowledge in their brains, thus reconstructing the entire physical three-dimensional world in their minds and grasping the properties of each object (e.g., "This is a person; they might move in the next second." "This is a table; it won't move unless pushed by an external force"). Well, autonomous driving perception systems do something similar: they need to reconstruct a highly understood, structured three-dimensional physical world from the limited signals captured by sensors within their model parameters. However, in this regard, model research faces a fundamental contradiction between the physical world and model computation: the mathematical computational basis of deep learning models is inherently incompatible with the spatial distribution patterns of the physical world.

1.1.1 The Dimensional Gap: Dense Matrices and Sparse Manifolds

The rapid advancement of modern computer vision is largely attributed to the perfect adaptation of convolutional neural networks to image data. From a mathematical perspective, a digital image is a dense matrix of size H × W, where pixel information is continuously distributed and semantic information is dense. This structure is inherently well-suited to the GPU's SIMD (Single Instruction Multiple Data) architecture, enabling large-scale matrix multiplication (GEMM) to run with extremely high throughput.

However, LiDAR perceives the 3D physical world very differently. It learns that an obstacle reflects its emitted laser by emitting laser light and receiving reflected laser light from the same direction. Its output—the point cloud—is an unordered, discrete set

of (x, y, z, r) coordinates. Therefore, the openness of the physical world results in extreme sparsity in the point cloud output by LiDAR in 3D coordinates. In classic autonomous driving datasets such as nuScenes (Caesar et al., 2020), effective point cloud data accounts for less than 5% of the 3D bounding box volume. This means that simply extending 2D convolution to 3D space (i.e., constructing a 3D matrix of $H \times W \times D$ for 3D convolution calculations) would not only result in an explosive increase in computational cost, but over 95% of computational resources would be wasted on multiplication operations between "air" and the convolution kernel weights (i.e., $0 \times W = 0$).

1.1.2 Evolution of Representation: From VoxelNet's Cube to PointPillars' Prism

To address the aforementioned problem of explosive computational cost, the academic community has undergone a series of evolutions in representation methods, which are essentially continuous trade-offs between resolution and computational cost.

Early VoxelNet (Zhou & Tuzel, 2018) attempted to preserve the complete geometric structure of 3D space by discretizing the space into tiny voxels. In this way, by controlling the size of the voxels, researchers could control the model's resolution, thereby adjusting the terrifyingly high computational cost of 3D convolution calculations. While this method retains the fine-grained features of the Z-axis compared to the later PointPillars (Lang et al., 2019), its computational complexity still increases cubically with increasing spatial resolution ($O(N^3)$). This means its computational load remains explosive. The resulting memory fragmentation and computational latency make it unsuitable for real-time inference on edge computing devices. The subsequent SECOND (Yan et al., 2018) introduced sparse convolution technology based on VoxelNet, using hash table indexes to allow the GPU to perform calculations only on non-empty voxels. This initially alleviated the computational bottleneck, increasing the speed of 3D convolution calculations by a staggering 3 to 4 times, but also increased the GPU's random-access requirements for memory bandwidth.

In computationally constrained industrial deployment scenarios, a more thorough dimensionality reduction strategy is needed. For example, PointPillars (Lang et al.,

2019) proposed an engineering solution based on "first principles." It proposes an observation: vehicles and surrounding objects (such as other vehicles and pedestrians) primarily move on a two-dimensional manifold of the ground, and the vast majority of objects to be perceived do not have complex overlapping structures along the Z-axis. Even if they do, it's not significant. Based on this physical prior, PointPillars performs "Bird's-Eye View Collapse," which infinitely stretches the 3D space along the Z-axis, dividing it into vertical pillars. This operation mathematically transforms the sparse 3D problem into a dense 2D pseudo-image problem. This transformation allows the model to directly avoid expensive 3D convolution computations, achieving detection using mature and highly optimized 2D CNN operators (such as ResNet and SSD). This operation not only enabled PointPillars to achieve astonishing detection speeds (62 Hz to 105 Hz) but also preserved the continuity of GPU memory access, allowing it to maximize information throughput with limited computing power. This is why this study chose PointPillars as the baseline model.

1.2 Problem Statement: The "Impossible Triangle" of Robust Perception

Although PointPillars solved the problem of "slow computation," a single-modal model is insufficient to truly build a highly robust autonomous driving perception system. Single-modal models, due to their singular perception source, are inevitably constrained by the dual bottlenecks of physical perception capability and data economy.

1.2.1 Geometric-Semantic Asymmetry

LiDAR is a geometric giant, perceiving the three-dimensional structure of the environment with centimeter-level precision. However, LiDAR is also a semantic dwarf, outputting point cloud data with only x, y, z three-dimensional coordinates and reflection intensity r, lacking semantic information. This often leads to difficulties in distinguishing between geometrically similar but different objects. In contrast, RGB images from a camera contain rich information such as color, texture, and text, but cannot be used to directly create 3D models of the surrounding environment. Modal fusion of the two seems inevitable. Therefore, the distillation training framework developed in this study inherently supports distillation training from multimodal

models to single-modal models.

1.2.2 The Economic Paradox of Supervised Learning

Deep learning methods are essentially data-driven algorithms, and their ultimate performance depends on the quality and quantity of labeled data. It's a classic case of "garbage in, garbage out." However, in the field of 3D object detection, there's a huge gap between the marginal cost of acquiring data and the marginal cost of labeling it.

1. Acquisition cost is extremely low: Autonomous driving test vehicles can collect terabytes of raw sensor data daily.

2. Labeling cost is extremely high: Human annotators need a series of alignment and calibration operations to label a 3D bounding box in 3D space, making labor costs extremely high.

This cost asymmetry leads to an awkward situation: the industry possesses massive amounts of raw sensor data, but only a small fraction of it is labeled. If the model learns only from labeled data, its performance ceiling is inevitably locked. Even increasing the labeling budget will be prohibitively expensive. Therefore, semi-supervised learning (SSL) is an essential path for models to achieve highly robust perception.

1.2.3 Violation of Physical Consistency: When Data Augmentation Violates Physical Laws

In semi-supervised learning, consistency regularization is a classic method to improve model robustness. It requires the model to maintain consistent predictions even after agitation. In the 2D image domain, MixUp (Zhang et al., 2017) is a standard augmentation technique. However, directly transferring MixUp to the 3D point cloud domain leads to "physical fallacies." 3D objects occupy exclusive physical space (a macroscopic manifestation of the Pauli Exclusion Principle). Simple point cloud overlay can result in two cars overlapping in space, or a tree inside a car, which clearly violates physical laws. Such "noisy samples" that violate physical laws disrupt the

model's learning of object geometric boundaries, making data augmentation counterproductive.

Therefore, LaserMix (Kong et al., 2023) proposes an augmentation method that respects the physical scanning characteristics of LiDAR sensors. It divides point cloud data into three layers (upper, middle, and lower) based on the LiDAR beam angle, allowing different scenes to exchange data from the upper, middle, or lower layers. Due to the significant differences in physical priors between different layers of data, the resulting hybrid scenes are physically plausible. This method cleverly enhances the data without compromising physical consistency, forcing the detector to learn the geometric features of objects in the middle layer without relying on contextual information from upper and lower layers.

1.3 The Proposed Framework

Based on the above analysis of computational efficiency, information entropy, and physical consistency, this study proposes a Teacher-Student architecture distillation training framework. This framework is based on the OpenPCDet platform (OpenPCDet Development Team, 2020) and aims to enable student models to robustly inherit the detect capabilities of teacher models.

The specific components of this distillation framework are as follows:

1. Cross-Architecture Knowledge Distillation: This study employs an efficient feature-level distillation method. This distillation method allows student models to learn the detection capabilities of heterogeneous detection head teacher models without significantly increasing computational costs during training. While the output model still uses a basic detection head in its architecture, its performance approaches that of a teacher model with a two-stage detection head.

2. Feature-Level Beam-Wise Mixing: Inspired by LaserMix (Kong et al., 2023), this study designs a BWM module. This module simulates LaserMix's approach to introduce adversarial interference into a portion of the distillation process, thereby forcing the student model to learn robustly rather than rote memorization of context.

3. Consistency Regularization: This study achieves the aforementioned adversarial interference by imposing consistency regularization constraints, forcing the model to focus on learning the geometric features of objects rather than the context for detection.

1.4 Thesis Structure

The subsequent chapters of this thesis are as follows:

Chapter 2: Literature Review

Chapter 3: Research Methodology

Chapter 4: the Design of Experiments

Chapter 5: Results and Discussion

Chapter 6: Conclusion

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

The core challenge of autonomous driving perception lies in how to reconstruct the geometric and semantic information of the 3D scene in the model's feature space based on high-dimensional, sparse, and unstructured LiDAR point clouds and dense RGB images. From a mathematical perspective, the abstract essence of this problem can be summarized as solving a nonlinear mapping $f_\theta: X \rightarrow Y$ from a heterogeneous observation space $X \in R^{(N \times D)}$ to a structured semantic space $Y \in R^{(M \times K)}$. This chapter will systematically review the development of autonomous driving perception from scratch, focusing on the theme of "the trade-off between computational efficiency and feature representation". This chapter will discuss four main topics: sparse convolution operator design, anchor-free detection head construction, cross-modal feature space unification, and semi-supervised consistency learning.

2.2 Structured Representation of Sparse Point Clouds

LiDAR point clouds are extremely sparse in 3D space (effective duty cycle is typically less than 1%). Directly applying standard 3D convolutions, as in 2D image recognition, results in a computational complexity of $O(L \cdot W \cdot H)$. Moreover, the vast majority of this computational power is consumed in multiplying zero-value voxels with weights. This was the common problem with early models like VoxNet (Zhou & Tuzel, 2018). Therefore, designing an efficient "feature encoder" that allows point cloud data to be efficiently computed on GPUs was a primary issue in early model research.

2.2.1 Sparse Convolution and Manifold Preservation

To address the computational redundancy caused by voxelization, SECOND (Yan et al., 2018) proposed a sparse convolution algorithm based on hash indexes. Its mathematical essence is to utilize the sparsity of point clouds to restrict convolution operations to non-empty manifolds. Let the input feature be V_{in} and the output feature be V_{out}. In

standard convolution, the convolution kernel K will calculate voxels at all positions in the space. The sparse convolution algorithm will first establish a mapping relationship (Rule Generation) between the input index P_{in} and the output index P_{out} through a hash table. Then, through this mapping relationship, the algorithm will restrict the convolution kernel K to only perform calculations on non-empty voxels. The feature calculation formula for the output position $u \in P_{out}$ is:

$$V_{out}(u) = \sum_{i \in \mathcal{K}} W_i \cdot V_{in}(u+i)$$

$$\text{s.t.} \quad u + i \in P_{in} \tag{2.1}$$

This algorithm successfully reduced the computational complexity from the volume of three-dimensional space $O(V)$ to the number of non-empty voxels $O(N)$, thereby significantly improving the inference and training speed of the model and becoming the standard for almost all subsequent models.

2.2.2 Dimensional Collapse: Pseudo-Image Coding

Unlike its predecessors, PointPillars (Lang et al., 2019) observed that "the Z-axis features of objects in ground vehicle driving scenes are low-frequency", and thus boldly proposed the Pillar Encoding method. This method discretizes the three-dimensional space into vertical pillars of $H \times W$. Then, a simplified version of PointNet feature aggregation is performed on the point set $P = p_i | i = 1 \dots N$ within the pillar:

$$V_{frustum}(u, v, d) = \alpha(u, v, d) \otimes F(u, v) \tag{2.2}$$

This operation collapses the sparse 3D point cloud $P \in R^{N \times 3}$ into a dense 2D pseudo-image $I_{pseudo} \in R^{C \times H \times W}$. Through this dimensionality reduction transformation, PointPillars can directly utilize highly optimized 2D CNN operators (such as ResNet (He et al., 2016)) in the 2D image domain to extract features from pillars, thus achieving its extremely high detection speed (62 Hz to 105 Hz). Even at the cost of sacrificing geometric information along the Z-axis, PointPillars' detection accuracy still reaches the level of state-of-the-art models of its time, achieving a dual success in both speed and accuracy.

2.3 Evolution of Detection Paradigms: From Geometric Aggregation to Anchor-Free Regression

Designing a good "feature encoder" is not enough; how to "decode" the geometric and semantic information of the object from the encoded "feature map" (i.e., the design of the detection head) is another major problem. Although the well-known anchor-based method has achieved great success in the field of 2D image recognition, its fixed preset bounding box design is difficult to adapt to the complex object poses in the field of 3D object detection.

2.3.1 Feature Interaction and Hierarchical Abstraction of Point-Voxel

To address the quantization error problem caused by voxelization, PV-RCNN (Shi et al., 2020) proposed a two-stage hybrid architecture that combines the efficiency of voxels with the geometric accuracy of the original point cloud. The core of this hybrid architecture lies in its original Voxel Set Abstraction (VSA) module. First, PV-RCNN uses Farthest Point Sampling (FPS) to sample key points k_j from the point cloud. Then, PV-RCNN performs ball query on these key points k_j and aggregates the "queried" features at multiple scales:

$$f(k_j) = \mathrm{MLP}\left(\bigcup (l = 1)^{L} \mathrm{Group}\left(V^{((l))}, k_j, r_l\right)\right) \tag{2.3}$$

This mathematical design constructs a non-uniform receptive field for the model using different r_l radii. Following this, PV-RCNN maps keypoint features back to the RoI region using RoI Grid Pooling, effectively correcting the localization bias from Voxel CNN. Through these designs, PV-RCNN perfectly combines the computational efficiency of voxels with the spatial accuracy of the original point cloud, establishing a performance upper bound for two-stage detectors.

2.3.2 Center of mass regression and rotation invariance

To address the difficulty of matching anchors to 3D objects with free rotation angles using anchor-based methods, CenterPoint (Yin et al., 2021) proposed a keypoint-based attribute regression method. Inspired by CenterNet (Zhou et al., 2019), CenterPoint abandoned the commonly used method of calculating loss based on anchor matching

IoU for object detection and instead adopted the method of predicting the target center (heatmap) to achieve object detection. For the true centroid p, CenterNet smooths the discrete true point prediction into a continuous probability distribution prediction through a two-dimensional Gaussian kernel $Y_{xy} = \exp\left(-\frac{(x-p_x)^2+(y-p_y)^2}{2\sigma^2}\right)$, thereby effectively alleviating the quantization noise caused by voxelization. In addition, to alleviate the problem of imbalance of object classes in the samples, CenterNet also introduced a variant Focal Loss to optimize the loss function:

$$L_{hm} = -\frac{1}{N}\sum_{xy}\begin{cases}\left(1-\backslash hatY_{xy}\right)^{\alpha}\log\left(\backslash hatY_{xy}\right) if\ Y_{xy}=1 \\ \left(1-Y_{xy}\right)^{\beta}\left(\backslash hatY_{xy}\right)^{\alpha}\log\left(1-\backslash hatY_{xy}\right) otherwise\end{cases} \tag{2.4}$$

Then, using a feature map with shared weights, CenterPoint directly regresses the geometric properties of the target object (length l, width w, height h, velocity v, etc.) based on the centroid's position. Addressing the mathematical singularity of the 3D bounding box rotation angle θ at the boundary $[0,2\pi]$ (i.e., the abrupt angle change problem), CenterNet achieves a continuous mapping of the rotation angle θ through a bivariate parametric regression design of $(\sin\theta, \cos\theta)$, thus significantly enhancing the model's generalization ability for detecting objects with arbitrary orientations. Through its decoupled detection design paradigm and efficient mathematical design, CenterPoint has successfully become the mainstream mathematical framework for current object detection model heads.

2.4 Geometric Alignment Through Multimodal Fusion: From "Hard Projection" to "Soft Dimensional Upgrading"

LiDAR point clouds are sparsely distributed in Cartesian space $X_{LiDAR} \in R^3$, while camera pixels are neatly and densely arranged in perspective space $X_{Cam} \in R^2$. To achieve cross-dimensional fusion of information from these two vastly different modalities, mathematically, one of their topological structures must be disrupted.

2.4.1 Hard projection: PointPainting

PointPainting (Vora et al., 2020) is the most representative pre-fusion method. It first uses a geometric projection matrix $P = K[R|t]$ to directly "paint" the 3D point p_i

onto the 2D image plane:

$$u_i = \frac{1}{z_i} K \cdot p_i \tag{2.5}$$

Then, it directly appends the output features $S(u_i)$ generated by the image segmentation network to the point cloud vector. Mathematically, this is a very straightforward and deterministic process, ensuring that each point cloud p_i contains its corresponding image features. However, the extremely sparse LiDAR point cloud does not correspond one-to-one with image pixels. An object represented by a pixel may not have been hit by the laser. This fusion method results in a significant amount of semantic density from the camera being discarded during modal fusion. Furthermore, the topological structure of image semantics is a frustum. This hard projection process compresses all object information along the entire ray onto a single point. If the depth estimation is slightly off, or if occlusion exists, PointPainting can easily "paint" the "tree" label from the background onto the point cloud of a car in the foreground. This fusion method, which adapts dense image information to sparse LiDAR point clouds through dimensionality reduction, achieves good performance in a minimalist way, but it also significantly wastes the camera's semantic information, limiting the model's performance ceiling.

2.4.2 Dimensional Upgrading and Probability Dispersion: BEVFusion

To address the shortcomings of PointPainting, BEVFusion (Liu et al., 2023) employs a completely opposite mathematical approach: Lifting. Since projecting a 3D manifold onto a 2D manifold results in information loss, just upgrade the 2D manifold to 3D. To achieve this, BEVFusion introduces the LSS (Lift, Splat, Shoot) operator, whose core mathematical operation is the Outer Product, which allows for dimensionality enhancement of the manifold. The model no longer predicts a single depth parameter d, but rather the probability distribution $\alpha \in R^D$ of the depth parameter d (e.g., D = 80 discrete depth intervals). The formula for calculating the view frustum feature V from the image feature $F \in R^C$ is as follows:

$$V_{\text{frustum}}(u, v, d) = \alpha(u, v, d) \otimes F(u, v) \tag{2.6}$$

This formula essentially "smears" 2D image features F outwards along the camera's line of sight. If the depth is uncertain (α is flat), then the image features will be uniformly diffused across the entire ray; if the depth is fixed, the features will cluster at a certain point on the ray. In this way, the image features formed by each pixel of the camera can be diffused or clustered along a dedicated ray in 3D space. No camera semantic information is discarded during the fusion process because it was "not hit by the laser".

Finally, BEVFusion uses raster pooling ($\sum V_{frustum}$) to physically metrically align the view frustum features and LiDAR features within the same BEV coordinate system, thus achieving dimensionality upscaling. Compared to the lossy projection of PointPainting (3D to 2D), BEVFusion's probabilistic projection (2D to 3D) mathematically solves the heterogeneous alignment problem between the two modalities without loss, thereby improving the performance ceiling of the multimodal fusion model.

2.5 Consistency Regularization

In semi-supervised learning, consistency regularization is crucial. It refers to designing a loss function that ensures the model's predictions are as consistent as possible with respect to a sample x and its perturbed version $\tilde{x}$. However, in 3D point cloud space, perturbation methods commonly used in 2D image processing often disrupt the geometry of objects and even cause point cloud features to violate physical laws.

2.5.1 Mean Teacher

Mean Teacher (Tarvainen & Valpola, 2017) is a well-known semi-supervised training architecture. It involves two models working together during training: a teacher model and a student model. The student model updates its parameters through backpropagation using a loss function, while the teacher model's parameters are the EMA (exponential moving average) of the student model's parameters:

$$\theta'_t = \alpha\theta'_{t-1} + (1-\alpha)\theta_t \tag{2.7}$$

This semi-supervised training architecture forces the student model to produce

predictions that are as consistent as possible with the teacher model through a consistency loss function, thus achieving consistency supervision of the student model.

2.5.2 Rigid body constraints in geometric space: SESS

To address the lack of effective perturbation methods in the field of 3D object detection, SESS (Zhao et al., 2020) points out that consistency regularization supervision in 3D detection must be based on the “Euclidean Group”. In 3D point clouds, blindly adding Gaussian noise to the input point cloud can easily lead to non-rigid deformation of the output 3D bounding box, which severely violates physical rules. Therefore, SESS introduces effective point cloud perturbation methods for 3D bounding boxes (such as global rotation and scaling), and designs an IoU-based loss function based on this:

$$L_{cons} = 1 - IoU\left(B_{teacher}, T^{-1}(B_{student})\right) \tag{2.8}$$

Here, T^{-1} refers to the inverse transform of the input transform.

2.5.3 Physical splicing of sensor manifolds: LaserMix

The MixUp method ($\lambda I_A + (1-\lambda) I_B$), used in 2D image processing to generate "ghosting" images, can lead to two vehicles overlapping on the same voxel if used directly in 3D point clouds ($P_{mix} = P_A \cup P_B$). This is physically impossible. Such perturbations that violate physical rules pollute the model's feature space, reducing its detection accuracy and ultimately proving counterproductive.

LaserMix (Kong et al., 2023) observed that LiDAR performs linear scanning based on pitch angle, and the distribution manifold of its point cloud is a series of concentric rings in spherical coordinates. Therefore, LaserMix abandoned the traditional MixUp method and instead used a spatial mask to mix point cloud scenes A and B:

$$\widetilde{P} = \{p \in P_A | \phi(p) \in R_A\} \cup \{p \in P_B | \phi(p) \in R_B\} \tag{2.9}$$

Where $\phi(p)$ is the pitch angle of the point.

This method of generating hybrid scenes is similar to piecing together the "upper half" of scene A and the "lower half" of scene B. From LiDAR's perspective, such a "hybrid scene" is physically possible (Valid Physical Scene). This ensures that the hybrid data,

as perturbation samples $\tilde{x}$, still exists on the real data manifold of the LiDAR point cloud in the physical world, rather than in an artificially created noisy manifold.

2.6 nuScenes Detection Score

The evaluation metric of the classic KITTI dataset is a pure geometric overlap metric based on IoU (AP_{3D}), lacking a time dimension. Therefore, the later nuScenes (Caesar et al., 2020) dataset mathematically reconstructed its metric function, proposing NDS (nuScenes Detection Score). The essence of NDS is a weighted sum of a set of decoupling error terms:

$$NDS = \frac{1}{10}\left[5 \cdot \mathrm{mAP} + \sum_{k \in \mathrm{TP}} (1 - \min(1, \mathrm{TP}_k))\right] \tag{2.10}$$

Where TP_k includes:

1. **ATE (Translation): Euclidean distance (meters) between the center points.**
2. **ASE (Scale): Volume error (1 - IoU of aligned boxes).**
3. **AOE (Orientation): Angular difference (radians), used for normalizing the cycle.**
4. **AVE (Velocity): Velocity vector error under the L2 norm.**
5. **AAE (Attribute): Attribute classification error.**

This design forces the model to not only draw accurate bounding boxes (geometrically), but also to accurately predict the motion state of objects (physically). This is precisely why object detection models based on the CenterPoint detector head can dominate the nuScenes rankings.

2.7 Conclusion: The Spiral Ascent of Computing Power, Geometry, and Physics

In conclusion, this study argues that the work of our predecessors is not merely a history of the development of autonomous driving object detection models, but also a history of human evolution in mathematically reconstructing existing methods of perceiving the physical world. From computational graph reconstruction using sparse operators to continuous geometric regression in the anchorless paradigm, from cross-modal attitude quantity unification of heterogeneous manifolds to semi-supervised generalization

driven by physical consistency, each step of development is a great testament to humanity's pursuit of the ideal of intelligent spatial perception.

CHAPTER 3

RESEARCH METHODOLOGY

3.1 Framework Overview

This study constructs a distillation training framework based on a Teacher-Student architecture. The aim of this study is to enable the student model to robustly inherit the geometric understanding capabilities of the teacher model without significantly increasing computational complexity.

Teacher (CenterPoint): CenterPoint (Zhou, Wang, et al., 2019) uses an anchorless detection head design and achieves detection through a two-stage calculation process (1. predicting the center point 2. regressing object attributes based on the center point). The NDS score in the NuScenes dataset is as high as 60.70.

Student (PointPillar): PointPillar (Lang et al., 2019) uses a traditional anchor-frame-based detection head, which has limited detection capabilities, with an NDS score between 45 and 50.

This study achieves feature-level distillation by designing a distillation loss $\mathcal{L}_{\mathcal{KD}}$ that forces the BEV feature space $\mathcal{F}_{\mathcal{S}}$ of Student model to fit to the feature space $\mathcal{F}_{\mathcal{T}}$ of the Teacher model.

Building upon this foundation and inspired by LaserMix (Kong et al., 2023), this study proposes the Feature-Level Beam-Wise Mixing (BWM) method. Based on the physical properties of LiDAR beams scanning row by row, this method treats the BEV feature map $\mathrm{F} \in \mathrm{R}^{\mathrm{C}\times\mathrm{H}\times\mathrm{W}}$ as a set of H physical scan rows. A BWM operator $\Phi(\mathrm{F_i}, \mathrm{F_j})$ is designed to perform row-wise interleaving on the feature manifold, thereby generating mixed features $\mathrm{F_{mix}}$. This method cleverly constructs physically plausible but semantically discontinuous mixed features $\mathrm{F_{mix}}$. Such mixed features $\mathrm{F_{mix}}$ create adversarial interference during training, hindering the student model's fitting of its feature space to the teacher. This adversarial interference forces the student model to abandon rote learning of the teacher's feature map knowledge during distillation and instead focus on robust learning of its perceptual patterns.

Distillation Loss Constraint: Minimize $\mathcal{L}_{\mathcal{KD}}$ to achieve feature-level knowledge distillation.

Consistency Loss Constraint: Minimize $\mathcal{L}_{cons}$ to achieve robust learning of perceptual patterns of teacher by student.

This study aims to achieve performance improvement of the PointPillar model trained under the above two constraints compared to the original PointPillar model. The goal is to give the classic PointPillar architecture the geometric intuition of CenterPoint and the perceptual robustness of LaserMix.

3.2 Cross-Architecture Knowledge Distillation

This study designed a Cross-Architecture Knowledge Distillation module. Its mathematical essence is to enable the feature space of Student (PointPillar) to fit to the feature map of Teacher (CenterPoint).

Let $\mathrm{F_S}, \mathrm{F_T} \in \mathrm{R}^{\mathrm{B\times C\times H\times W}}$ be the BEV feature tensors of Student and Teacher, respectively. To encourage the internalization of Teacher's spatial geometric intuition into Student's feature map, this study uses mean squared error (MSE) to define the topological projection loss $\mathcal{L}_{\mathcal{KD}}$:

$$\mathcal{L}_{\mathcal{KD}} = |\mathrm{F_S} - \mathrm{Detach}(\mathrm{F_T})|_2^2 \tag{3.1}$$

The Detach operation blocks gradient backpropagation to the Teacher.

3.3 Feature-Level Beam-Wise Mixing

Inspired by LaserMix (Kong et al., 2023), this study designed a Feature-Level BWM module. This module creates an adversarial interference to the process of the student learning the teacher's feature map through row-wise interleaving of feature maps.

Let the BWM operator be $\Phi(\mathrm{F})$. Let the binary feature map mask be $\mathrm{M} \in \{0,1\}^{\mathrm{H\times W}}$. For the feature F in the current batch, let the interfering feature along the batch be $\mathrm{F_{noise}} = \mathrm{Roll}(\mathrm{F}, \mathrm{shifts} = -1)$.

Therefore, this study defines the formula for calculating the mixture feature $\mathrm{F_{mix}}$ as follows:

$$\mathrm{F_{mix}} = \Phi(\mathrm{F}) = M \odot F + (1 - \mathrm{M}) \odot \mathrm{F_{noise}} \tag{3.2}$$

3.4 Consistency Regularization

This study applies a consistency regularization constraint $\mathcal{L}_{cons}$ to the student model during distillation training.

Let $\Phi(\mathrm{F_T})$ be the mixture feature of the teacher model and $\Phi(\mathrm{F_S})$ be the mixture feature of the student model. This study uses the mean squared error (MSE) of $\Phi(\mathrm{F_T})$ and $\Phi(\mathrm{F_S})$ to define the consistency loss:

$$\mathcal{L}_{cons} = |\Phi(\mathrm{F_S}) - \mathrm{Detach}\left(\Phi(\mathrm{F_T})\right)|_2^2 \tag{3.3}$$

3.5 Objective Function

Ultimately, the total loss function $\mathcal{L}_{total}$ consists of three parts: supervision loss, distillation loss, and consistency loss:

$$\mathcal{L}_{total} = \mathcal{L}_{task} + \lambda_{\mathrm{KD}} \cdot \mathcal{L}_{\mathcal{KD}} + \lambda_{\mathrm{cons}} \cdot \mathcal{L}_{cons} \tag{3.4}$$

1. Detection Task Loss ($\mathcal{L}_{task}$): The basic supervised loss of PointPillar, including classification loss (Focal Loss), bounding box regression loss (Smooth L1 Loss), and orientation classification loss (Softmax Cross-Entropy):

$$\mathcal{L}_{task} = \mathcal{L}_{cls} + \beta_{\mathrm{loc}}\mathcal{L}_{loc} + \beta_{\mathrm{dir}}\mathcal{L}_{dir} \tag{3.5}$$

2. Distillation Loss ($\mathcal{L}_{\mathcal{KD}}$): Responsible for gradient backpropagation of the semantic distillation loss across architectures, the equation is:

$$\mathcal{L}_{\mathcal{KD}} = |\mathrm{F_S} - \mathrm{Detach}(\mathrm{F_T})|_2^2 \tag{3.6}$$

3. Consistency Loss ($\mathcal{L}_{cons}$): Responsible for the gradient backpropagation of the

consistency regularization loss, the equation is:

$$\mathrm{F}_{\mathrm{mix}} = \Phi(\mathrm{F}) = M \odot F + (1 - \mathrm{M}) \odot \mathrm{F}_{\mathrm{noise}} \tag{3.7}$$

CHAPTER 4

THE DESIGN OF EXPERIMENTS

4.1 Dataset and Evaluation Metrics

This study uses the NuScenes (Caesar et al., 2020) full dataset as the basic dataset for the experiments. This dataset contains 1000 scenes (20 seconds each) from two cities, Boston and Singapore. Furthermore, 3D bounding boxes are annotated for 23 object categories (such as cars, pedestrians, bicycles, trucks, etc.) across 40,000 keyframes. The decompressed size of this dataset is approximately 403GB.

Meanwhile, this study uses standard mAP and NuScenes' innovative NDS as the main evaluation metrics for model performance.

4.2 Experimental Environment

The models were trained in a containerized Ubuntu 22.04 environment using the OpenPCDet framework based on Python 3.10, PyTorch 2.1.2, and CUDA 11.8. The hardware configuration consisted of an Intel(R) Xeon(R) Platinum 8481C CPU (16 vCPUs), a single NVIDIA RTX 4090D GPU (24GB), 80GB of RAM, and 575GB of total storage.

4.3 Experimental Plan

Exp 1 Baseline: Student (PointPillar)

First, this study will perform 10 epochs of PointPillar training. The trained model will serve as the baseline model for comparison with other models to evaluate performance improvements. At this stage, the model's loss function is simply: $\mathcal{L}_{task}$. Because the 4090D graphics card has ample video memory (24GB), the batch size can be set to 8 to increase training speed and allow the graphics card to remain fully utilized for a longer period.

Exp 2 Distillation: Student (PointPillar) + Teacher (CenterPoint)

Next, this study will perform 10 epochs of distillation training. CenterPoint is the

teacher model, and Pointpillar is the student model. During training, the parameters of the teacher model will be frozen, and only the parameters of the student model can be updated. At this time, the model's loss function consists of the supervised loss and the distillation loss: $\mathcal{L}_{task} + \lambda_{\mathrm{KD}}\mathcal{L}_{\mathcal{KD}}$, set $\lambda_{\mathrm{KD}} = 10.0$. Because the teacher model is now added to the GPU memory, the Batchsize needs to be set to 6 to ensure that the training process will not be interrupted due to insufficient GPU memory (caused by continuously reading a large number of complex point cloud scenes).

Exp 3 Distillation+BWM: Student (PointPillar) + Teacher (CenterPoint) + BWM

Finally, this study will conduct 10 epochs of distillation training enhanced with BWM. The newly added BWM module will introduce adversarial interference into the distillation process, forcing the student model to learn local geometric features invariant to the interference, thereby enhancing the model's robustness. The model's loss function at this point consists of the supervised loss, distillation loss, and consistency loss: $\mathcal{L}_{task} + \lambda_{\mathrm{KD}}\mathcal{L}_{\mathcal{KD}} + \lambda_{\mathrm{cons}}\mathcal{L}_{cons}$, set $\lambda_{KD} = 10.0$, $\lambda_{cons} = 5.0$. The batch size is also set to 6 to ensure sufficient GPU memory.

CHAPTER 5

RESULTS AND DISCUSSIONS

5.1 Experimental Results

After a total of more than 30 hours of training, the experimental results are as follows:

Metric	**Baseline (Exp1)**	**Distillation (Exp2)**	**Distillation +BWM (Exp3)**	**Teacher (CenterPoint)**
mAP ↑	0.3461	0.3994	0.4070	0.5003
NDS ↑	0.4951	0.5391	0.5468	0.6071
mATE (Translation) ↓	0.3843	0.3721	0.3670	0.3113
mASE (Scale) ↓	0.2695	0.2663	0.2625	0.2604
mAOE (Orientation) ↓	0.4295	0.4257	0.3884	0.4288

Metric	**Baseline (Exp1)**	**Distillation (Exp2)**	**Distillation +BWM (Exp3)**	**Teacher (CenterPoint)**
mAVE (Velocity) ↓	0.4947	0.3444	0.3504	0.2389
mAAE (Attribute) ↓	0.2009	0.1973	0.1987	0.1914

Here, mAP and NDS are indicators of the overall accuracy of the model in the object detection task. mATE, mASE, mAOE, mAVE, and mAAE are indicators of the model's accuracy in five aspects: Translation, Scale, Orientation, Velocity, and Attribute, respectively. Furthermore, this paper also introduces the CenterPoint score for comparison.

5.2 Discussions

1. mAP (Mean Average Precision)

The PointPillar trained by distillation showed a significant improvement over the PointPillar trained by normal methods. The addition of BWM further improved the model's performance, but it still lags considerably behind the performance of the teacher model. This indicates that distillation and BWM are effective in improving mAP scores, but due to limitations in model architecture (anchor-based detector head) and the number of training epochs (10 epochs), they cannot fully reach the level of the teacher model.

2. NDS (NuScenes Detection Score)

NDS is nuScenes' innovative comprehensive model performance evaluation metric. Similar to mAP, distillation significantly improves PointPillar's performance under the same training conditions. The addition of BWM further enhances its model performance, but it still lags behind teacher-level performance.

3. mATE (Mean Average Translation Error)

mATE is a strength of the teacher model, CenterPoint, due to its use of a center-based detector head. Distillation and BWM both reduce PointPillar's mATE, but it still lags significantly behind the teacher model.

4. mASE (Mean Average Scale Error)

The teacher model's mASE error has slight advantages compared to the baseline. Distillation and BWM both reduced the pointpillar's mASE error to some extent (albeit slightly). This indicates that distillation and BWM are both helping the pointpillar model to move closer to the teacher model.

5. mAOE (Mean Average Orientation Error)

The teacher model's capabilities on mAOE are similar to the baseline (pointpillar), so

the performance of the distilled model is almost identical to the baseline. However, the addition of BWM significantly reduces the mAOE error. This indicates that BWM allows the model to internalize a robust orientation-aware capability similar to LaserMix (Kong et al., 2023).

6. mAVE (Mean Average Velocity Error)

The teacher model exhibits a significant advantage over the baseline in mAVE. This is due to its center-based detection head architecture, which greatly simplifies the speed detection problem, resulting in a substantial advantage. Therefore, the distillation operation also significantly improves the pointpillar's performance on mAVE. While the addition of BWM slightly increases the maAVE error, this increase remains within acceptable limits.

7. mAAE (Mean Average Attribute Error)

The teacher model did not show a significant advantage in mAAE. Distillation slightly reduced the mAAE of Pointpillar, while BWM slightly increased it, but both were within the error range and could not be considered significant.

CHAPTER 6

CONCLUSION

Based on the experimental results presented above, we can see that the distillation training framework proposed in this study not only allows the student model to internalize the perceptual performance of the teacher model (especially the perception of speed), achieving a significant improvement in the NDS score (0.4951 → 0.5391), but the addition of its BWM module also allows the student model to improve its orientation perception under the pressure of interference similar to LaserMix (Kong et al., 2023), further improving the NDS score (0.5391 → 0.5468). This proves that the distillation loss constraint and consistency loss constraint we added to PointPillar's training are effective in improving the model's perceptual ability. Furthermore, these two loss constraints do not significantly increase the computational cost of model training. Affected by fluctuations in GPU temperature and GPU memory usage, the training times for Exp 1, Exp 2, and Exp 3 are all around 10 hours, with no significant difference. This demonstrates that the two additional constraints we added to the training significantly improve the perceptual performance of PointPillar without significantly increasing the computational cost.

Looking ahead, this paper believes that distillation technology has great potential in enabling simple architecture models to inherit the perceptual performance of complex architecture models. In the future, we might be able to distill the complex neural network structure of a sophisticated yet powerful perception model (where knowledge is stored in a complex structure, similar to structured theory) into the weights of a simpler architecture model (where knowledge is stored in a flattened manner, similar to experience). This would allow us to construct a powerful, lightweight object detection model, enabling autonomous vehicles to achieve high-performance, high-frame-rate object detection capabilities with limited computing power.

Finally, this paper expresses special thanks to the AutoDL (AutoDL, n.d.) platform for

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